

Lab 5: Data Visualization with ggplot

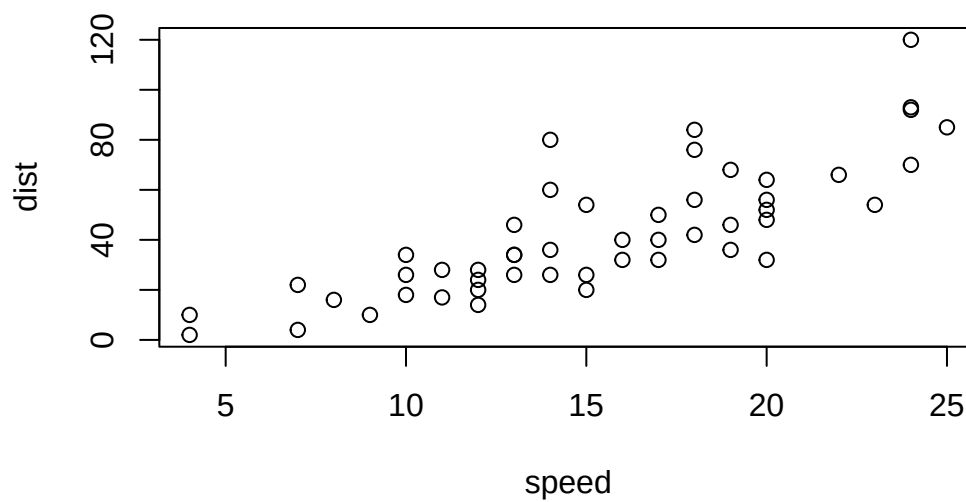
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Today we are exploring the **ggplot** package and how to use it for data visualization. Other ways in which figures and plots can be made in R are:

- “base” R
- other such add-on packages

“base” R plot

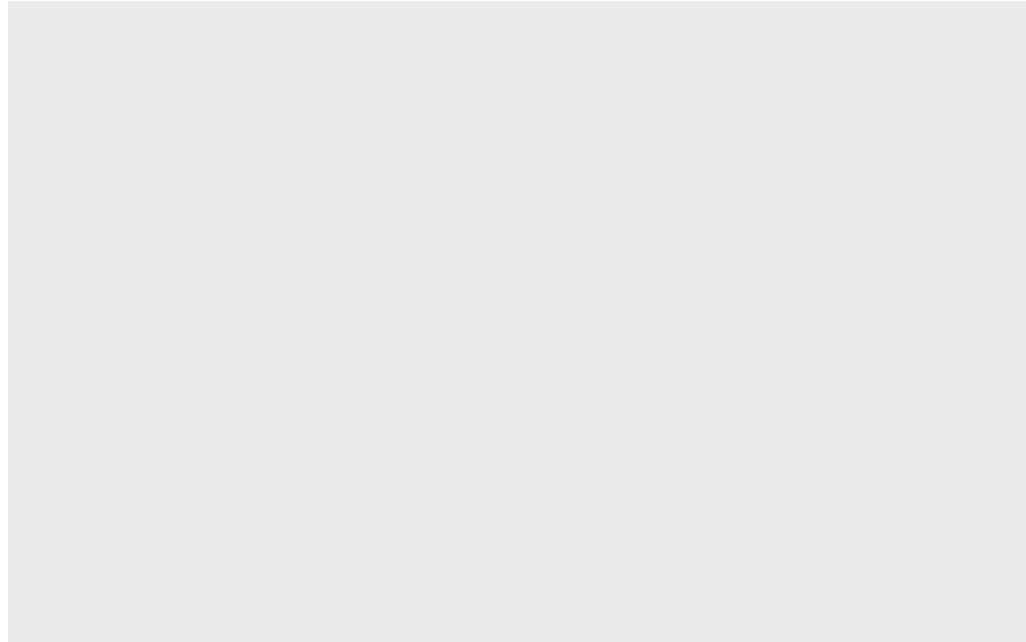
```
plot(cars)
```



Key point: “base” R is quick but not aesthetic

ggplot

```
library(ggplot2)  
ggplot(cars)
```

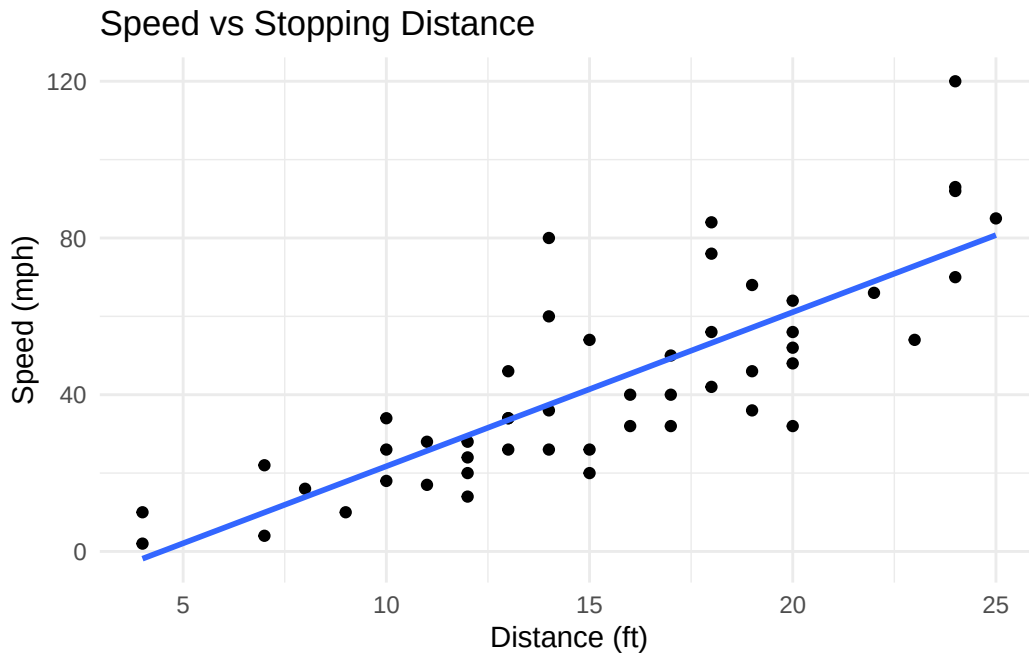


Every ggplot is composed of at least 3 layers:

- data (df with everything we want to plot)
- aesthetics
- geometry (includes the types of plot i.e. dot, column, etc.)

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method=lm, se=FALSE) +  
  labs(title="Speed vs Stopping Distance", x="Distance (ft)", y="Speed (mph)") +  
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



Key point: For simple “canned” graphs, “base” R is quicker and the code is more concise. As things get more elaborate, ggplot is better to make custom graphs with.

More elaborate plots

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

How many genes are in this dataset?

5196

How many upregulated genes are there?

127

```
nrow(genes)
```

```
[1] 5196
```

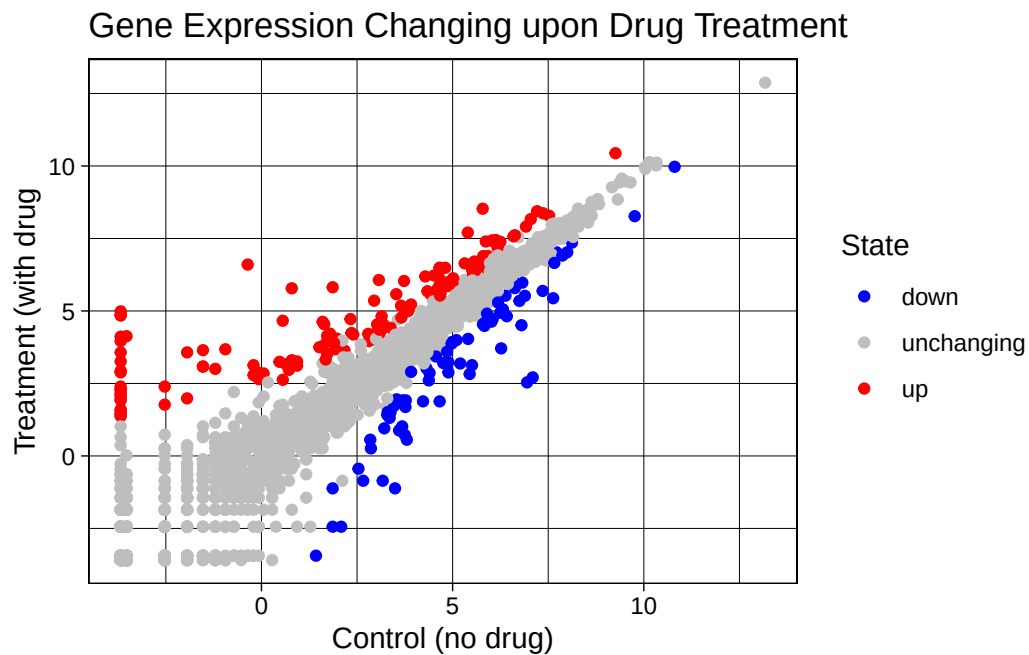
```
sum(genes$State=="up")
```

```
[1] 127
```

```
table(genes$State)
```

down	unchanging	up
72	4997	127

```
ggplot(data=genes) +  
  aes(x=Condition1, y=Condition2, col=State) +  
  geom_point() +  
  scale_colour_manual( values=c("blue","gray","red") ) +  
  labs(title="Gene Expression Changing upon Drug Treatment", x="Control (no drug)", y="Treatment (with drug)") +  
  theme_linedraw()
```



More Plotting

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"
gapminder <- read.delim(url)
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

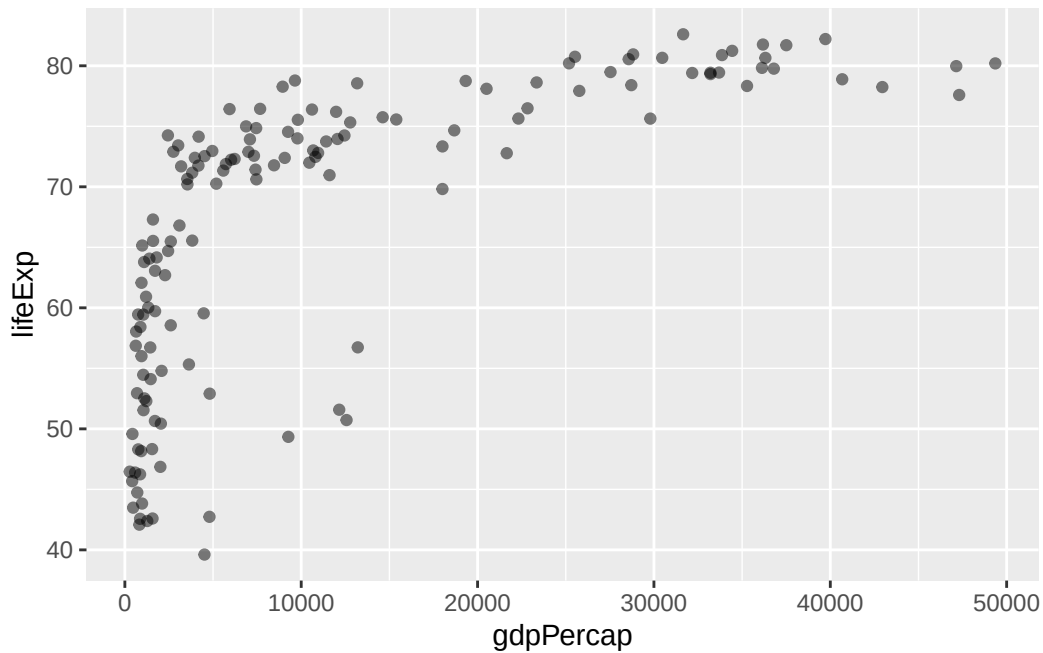
intersect, setdiff, setequal, union

```
gapminder_2007 <- gapminder %>% filter(year==2007)
head(gapminder_2007)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	2007	43.828	31889923	974.5803
2	Albania	Europe	2007	76.423	3600523	5937.0295
3	Algeria	Africa	2007	72.301	33333216	6223.3675
4	Angola	Africa	2007	42.731	12420476	4797.2313
5	Argentina	Americas	2007	75.320	40301927	12779.3796
6	Australia	Oceania	2007	81.235	20434176	34435.3674

To plot it,

```
ggplot(gapminder_2007) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point(alpha=0.5)
```



```
# alpha makes the dots more transparent
```

How many countries are in this dataset?

```
142
```

How many continents are in this dataset?

```
5
```

```
length(table(gapminder$country))
```

```
[1] 142
```

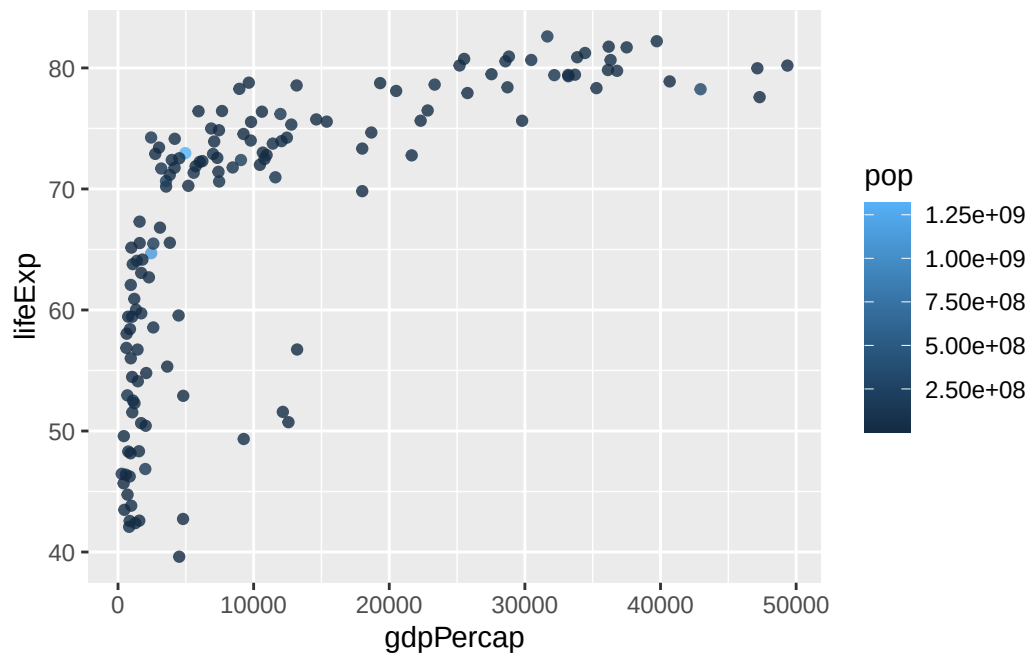
```
length(table(gapminder$continent))
```

```
[1] 5
```

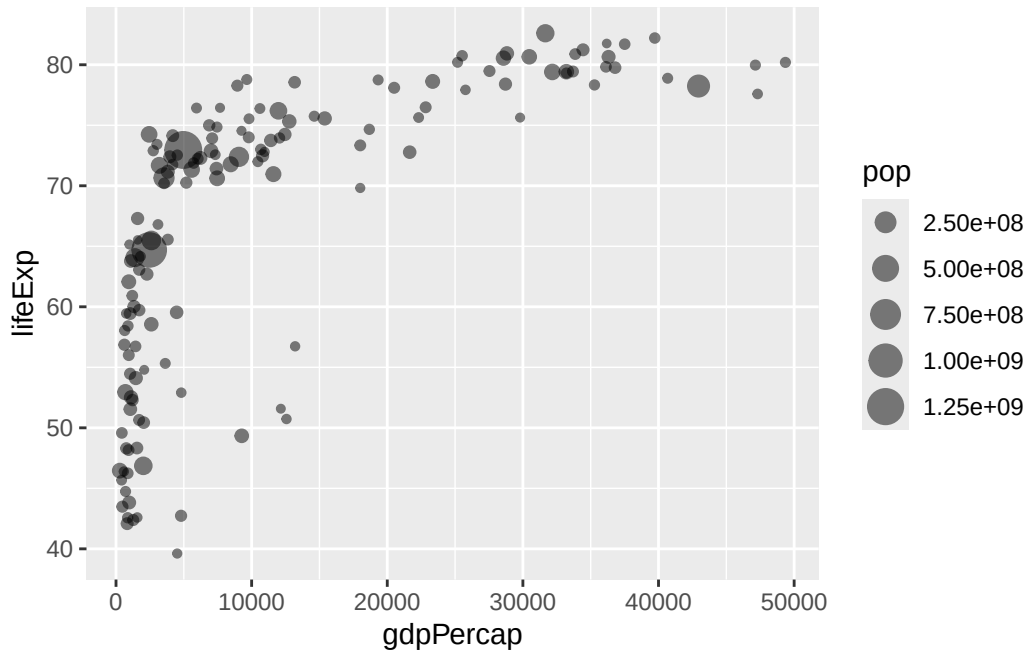
```
ggplot(gapminder_2007) +  
  aes(x=gdpPercap, y=lifeExp, color=continent, size=pop) +  
  geom_point(alpha=0.5)
```



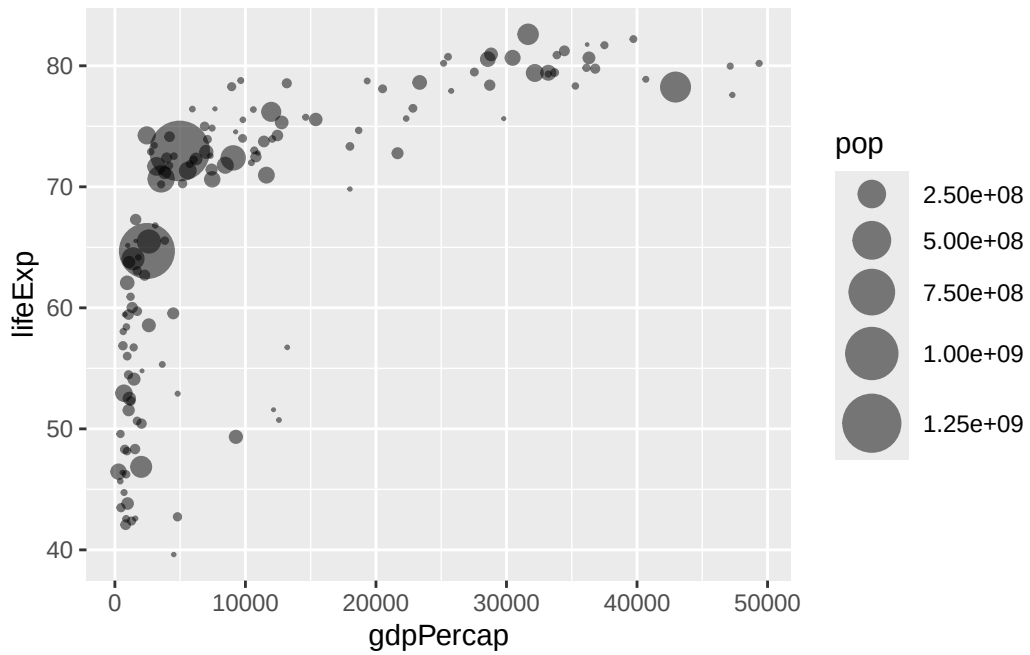
```
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, color = pop) +
  geom_point(alpha=0.8)
```



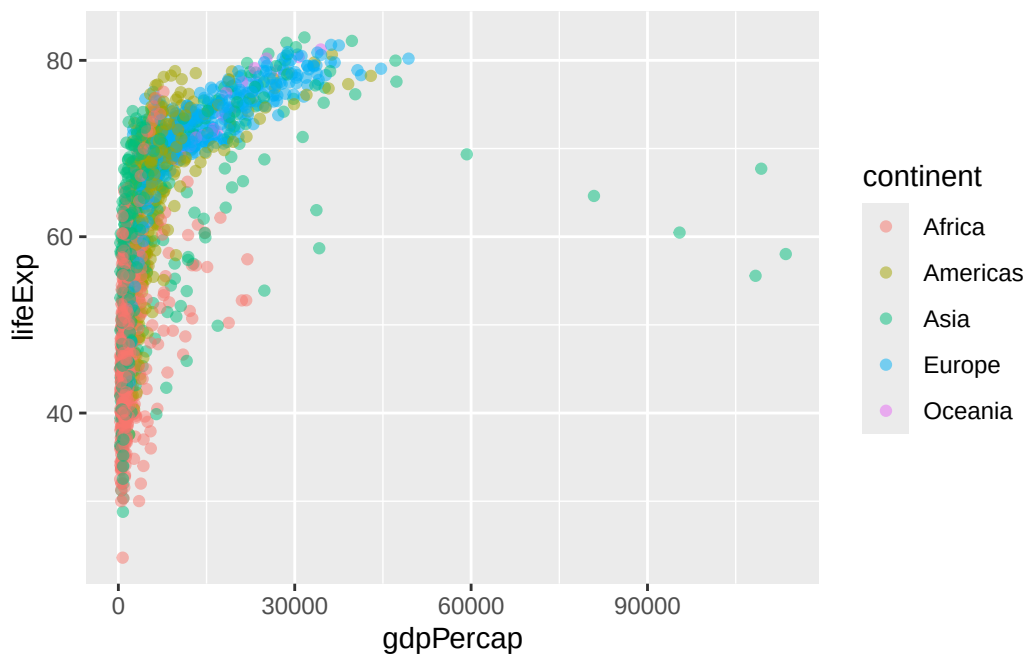
```
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, size = pop) +
  geom_point(alpha=0.5)
```



```
ggplot(gapminder_2007) +
  geom_point(aes(x = gdpPercap, y = lifeExp,
                 size = pop), alpha=0.5) +
  scale_size_area(max_size = 10)
```

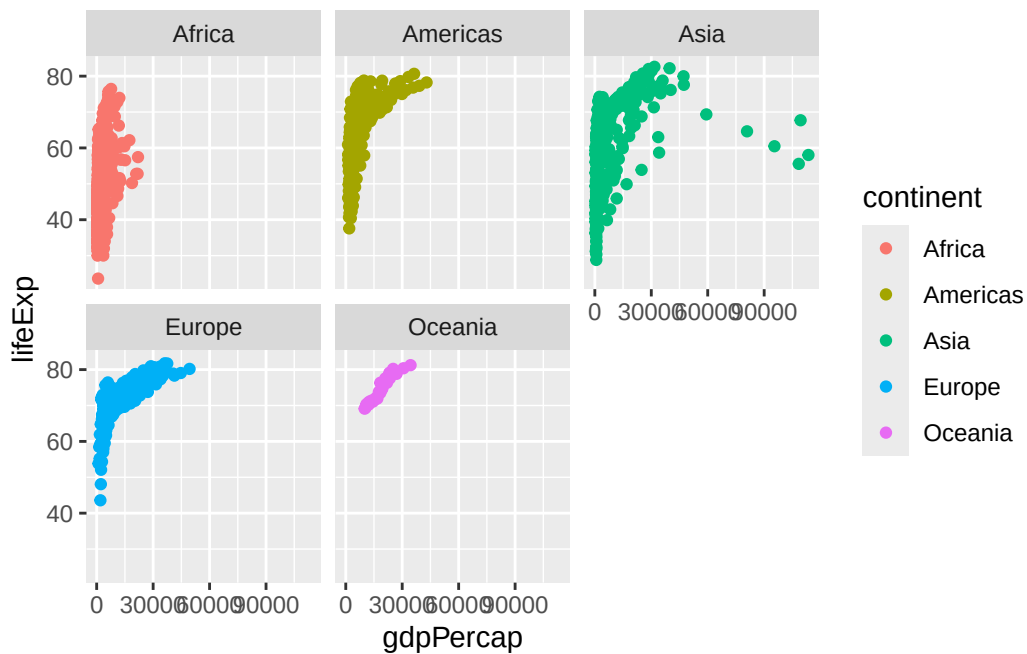
```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point(alpha=0.5)
```



```
kuw <- gapminder %>% filter(country=="Kuwait")
kuw
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Kuwait	Asia	1952	55.565	160000	108382.35
2	Kuwait	Asia	1957	58.033	212846	113523.13
3	Kuwait	Asia	1962	60.470	358266	95458.11
4	Kuwait	Asia	1967	64.624	575003	80894.88
5	Kuwait	Asia	1972	67.712	841934	109347.87
6	Kuwait	Asia	1977	69.343	1140357	59265.48
7	Kuwait	Asia	1982	71.309	1497494	31354.04
8	Kuwait	Asia	1987	74.174	1891487	28118.43
9	Kuwait	Asia	1992	75.190	1418095	34932.92
10	Kuwait	Asia	1997	76.156	1765345	40300.62
11	Kuwait	Asia	2002	76.904	2111561	35110.11
12	Kuwait	Asia	2007	77.588	2505559	47306.99

```
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent, label = country) +
  geom_point() +
  facet_wrap(~continent)
```



```
library(ggrepel)
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent, label = country) +
  geom_point() +
  geom_text_repel() +
  facet_wrap(~continent)
```

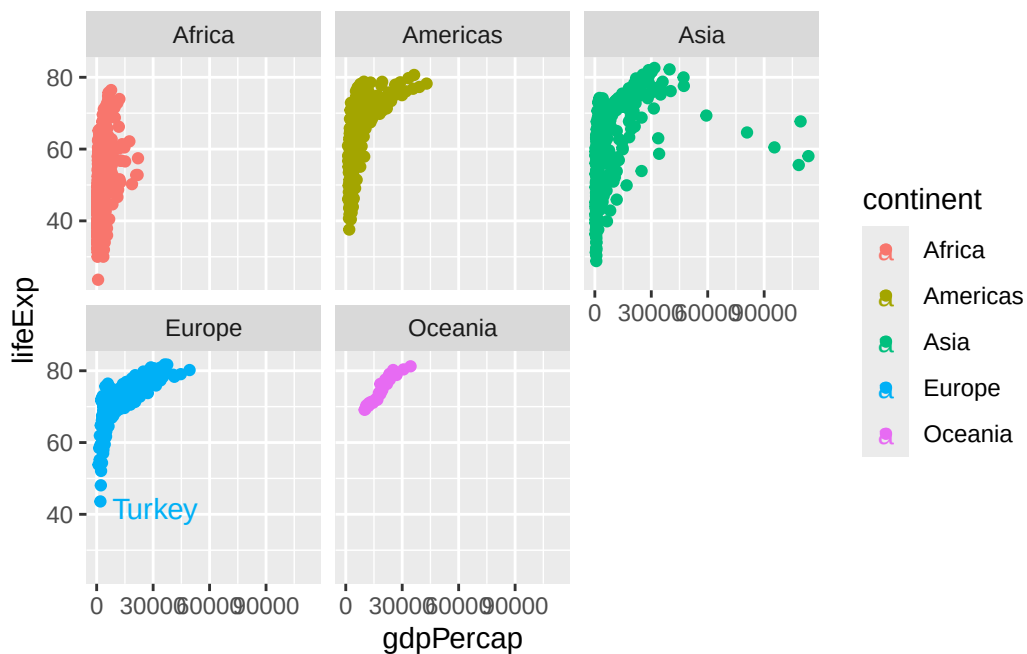
Warning: ggrepel: 624 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 359 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 300 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 24 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 396 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Below are some advantages of ggplot over base R:

1. **Layered Grammar of Graphics:** ggplot uses a consistent, layered approach. You build plots by adding layers (data, aesthetics, geoms, themes) with the `+` operator. This makes complex plots easier to construct and modify, compared to base R, where each plot type often requires different functions and arguments [1], [3], [2], [4], [5].
2. **Declarative Syntax:** You specify *what* you want to show (e.g., which variables map to axes, color, shape), not *how* to draw it. This makes code more readable and easier to maintain [1], [3], [2], [4], [5].
3. **Beautiful Defaults:** ggplot produces publication-quality figures with sensible defaults, so your plots look good without extensive tweaking. Base R gives you full control, but making plots look polished can be time-consuming [1], [3], [2], [4], [5].
4. **Faceting and Customization:** ggplot makes it easy to split data into multiple panels (facets) and customize aesthetics, legends, and themes. These features are more cumbersome in base R [3], [2].
5. **Extensibility:** ggplot is part of a large ecosystem of packages for advanced visualizations and customizations, making it more flexible for scientific work [1], [3], [2], [5].